

Francina Godwin Pali

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Work Experience

Outlier.AI – Remote

St. Louis, MO

Frontend Engineer & GenAI/LLM Model Trainer

Mar 2025 – Present

- Refined generative AI models through multi-turn prompt engineering to produce client-ready code.
- Guided AI toward secure coding practices: environment variables, bcrypt for password hashing, Knex.js transactions for atomic PostgreSQL updates.
- Optimized rendering performance and ensured cross-browser compatibility while implementing frontend tasks.
- Tech: JavaScript, Node.js, Express.js, PostgreSQL, React.js, HTML, CSS, Git.

Aidos, LLC – Ludus Lab

St. Louis, MO

Product Developer

Feb 2025 - Present

- Refactored codebase to eliminate redundancy and improve maintainability.
- Upgraded incompatible NPM packages and integrated alternatives to ensure compatibility.

Saint Louis University, School of Medicine

St. Louis, MO

Research Assistant

Oct 2024– Feb 2025

- Conducted predictive time series analysis using ARIMA models to forecast long-term healthcare expenditure.

Saint Louis University

St. Louis, MO

Teaching Assistant - Software Engineering

Aug 2024 – Dec 2024

- Led labs on Git (DAG, rebasing, merges), design patterns, CI/CD pipelines (YAML), and SOLID principles for web projects.

Saint Louis University

St. Louis, MO

Teaching Assistant – Applied Machine Learning

Jan 2024 – May 2024

- Evaluated and graded 50+ student assignments, ensuring comprehensive feedback to improve student performance.

Projects

Secure Task Management System

Backend Engineer

(In Progress)

- Full-stack web app for secure user task management with JWT authentication and role-based authorization.
- Implemented PostgreSQL joins and indexes for optimized queries across users and tasks.
- Handled multiple content types: JSON, URL-encoded, multipart (file uploads).
- Integrated external APIs for email notifications and file storage.
- Tech Stack: Node.js, Express.js, PostgreSQL, Knex.js, React.js, JWT, bcrypt.

Real-Time Face Detection

Software Engineer

Aug 2025 – Sep 2025

- Developed responsive web app with Node.js, Express, PostgreSQL, and React.js.
- Integrated an image recognition API to detect and highlight faces in uploaded images.
- Deployed frontend and backend services on Heroku with a managed Heroku Postgres add-on.

Soil Moisture and Temperature Forecasting

Machine Learning Engineer

Mar 2023 – May 2023

- Applied LSTM and customized Linear Regression models to forecast soil moisture and temperature with near-zero prediction error.
- Built interactive Tableau dashboard for precision agriculture insights.

Technical Skills

- **Web Development:** REST APIs (JSON, URL-encoded, multipart), Web Scraping (HTML parsing)
- **Databases:** PostgreSQL, MySQL
- **Programming/Frameworks:** JavaScript (ES6+), Python, Node.js, Express.js, React.js, Flask, PHP, Java, C, HTML5, CSS3
- **Debugging & Testing:** Postman, Chrome DevTools (Network, Console, Elements)
- **Data Analytics:** Data Analytics & Predictive Modeling

- **Core CS:** Data Structures & Algorithms (Recursion, Dynamic Programming, Trees, Graphs)
- **Machine Learning:** Applied ML
- **Cryptography**
- **Agile:** Scrum, Kanban | **Version Control:** Git/GitHub | **Cloud & Deployment:** Heroku | **Tools:** Tableau
- **CMS:** WordPress, Weebly

Publications

Energy Transitions over Five Decades: A Statistical Perspective on Global Energy Trends – Computers (MDPI)
May 2025 <https://doi.org/10.3390/computers14050190>

Forecasting World Health Expenditures: A Hybrid AI and Generative AI Approach – AIH (ACCSCIENCE Publishing)
Sep 2025 https://accscience.com/journal/AIH/articles/online_first/5617

Education

Saint Louis University	St. Louis, MO
MS in Computer Science (GPA 3.95 / 4.0)	Graduation Date: Dec 2024
Don Bosco Institute of Technology	Mumbai, India
BE in Information Technology (GPA 8.26/ 10)	Graduation Date: Aug 2022